

Acids & Bases in Organic Chemistry

Common definitions of acids and bases:

Arrhenius: Acids produce H^+ & bases produce OH^- ion in aqueous solution.

introduced in 1887
by the Swedish scientist [Svante Arrhenius](#)

Arrhenius acid increases the concentration of hydrogen ions in an aqueous solution

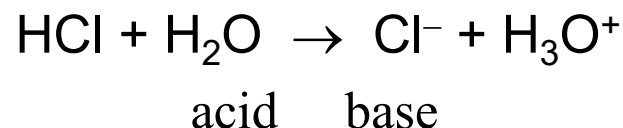


Arrhenius base increases the concentration of hydroxide ions in an aqueous solution



Brønsted-Lowry: Acids are H^+ donors & bases are proton acceptors.

developed by [Johannes Nicolaus Brønsted](#)
and [Thomas Martin Lowry](#)
independently in 1923



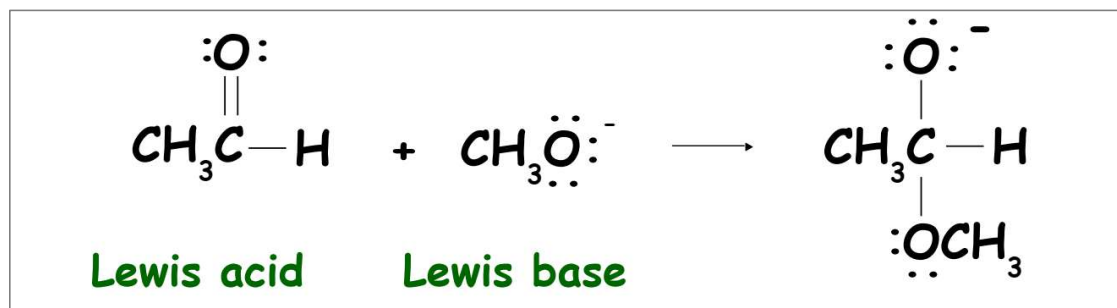
- Many of the organic reactions you will study involve acid-base reactions.

Understanding these reactions will require you to:

- Recognize organic compounds that can serve as acids or bases.
- Throw away the idea that anything with an OH is a base and anything with an H is an acid!!!

Acids & Bases in Organic Chemistry

The following reaction does not look like a “classic” acid-base reaction...neither reactant gains or loses an H^+ .



It is, however, a Lewis acid-base reaction.

Lewis acid-base concept

Gilbert N. Lewis

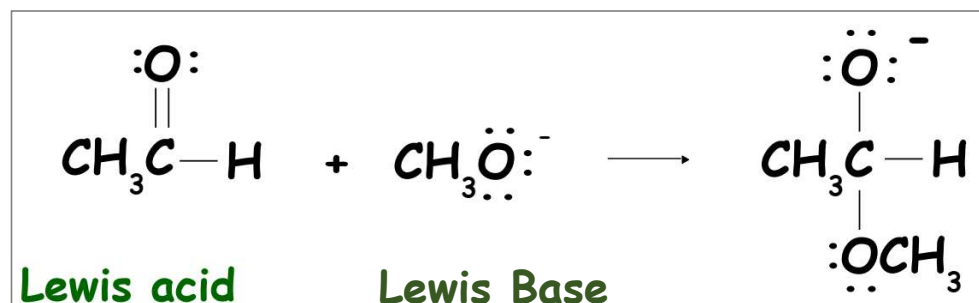
The Lewis acid-base definition is the broadest definition of acids and bases.

Lewis acid:

- an electron pair acceptor
- an **electrophile**
- “electron lover”
- a substance that accepts a pair of electrons to form a new bond

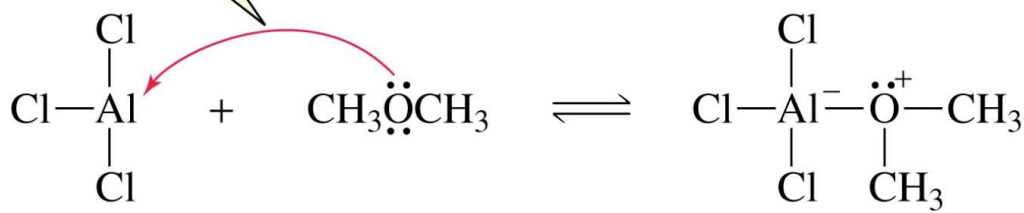
Lewis Base:

- an electron pair donor
- a **nucleophile**
- “nuclei lover”
- a substance with a pair of electrons that can be donated to another nucleus to form a new bond

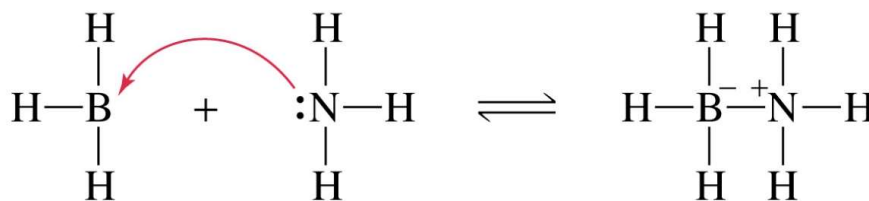


Example of Lewis acid-base

the curved arrow indicates
where the pair of electrons
starts from and where it ends up



aluminum trichloride dimethyl ether
a Lewis acid a Lewis base

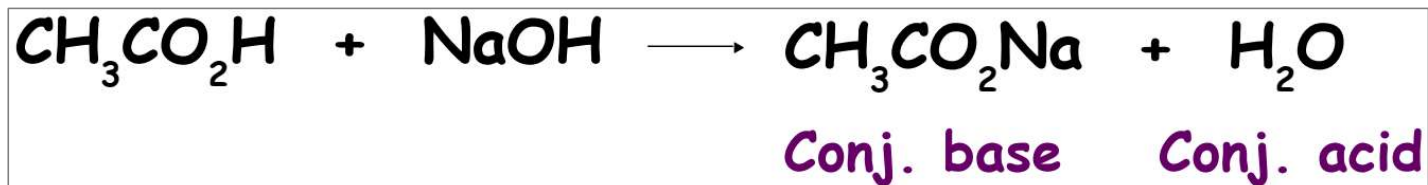


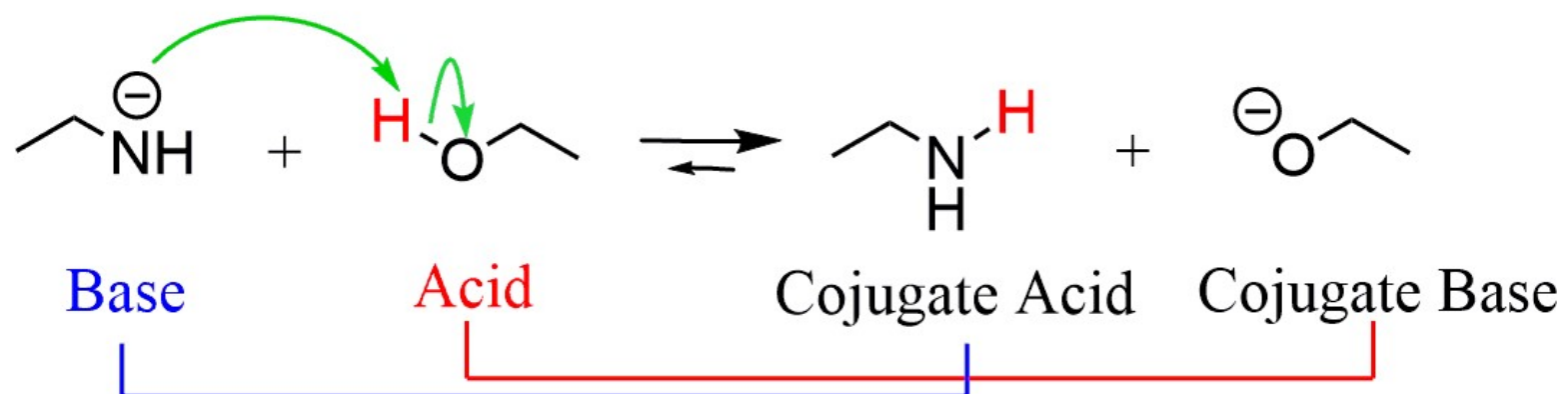
borane ammonia
a Lewis acid a Lewis base

Acid-base reactions

Acid-base reactions always produce a new acid and a new base.

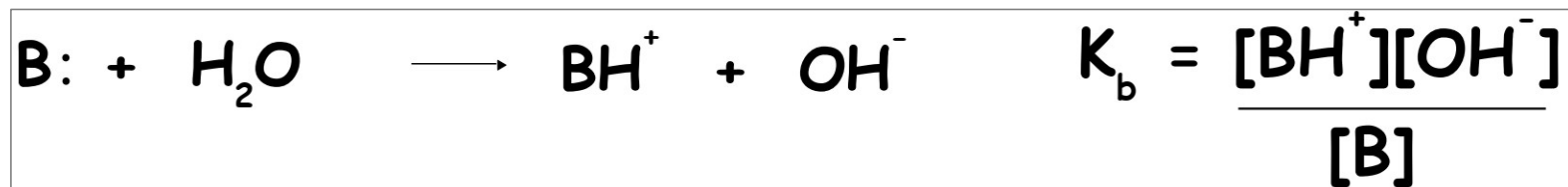
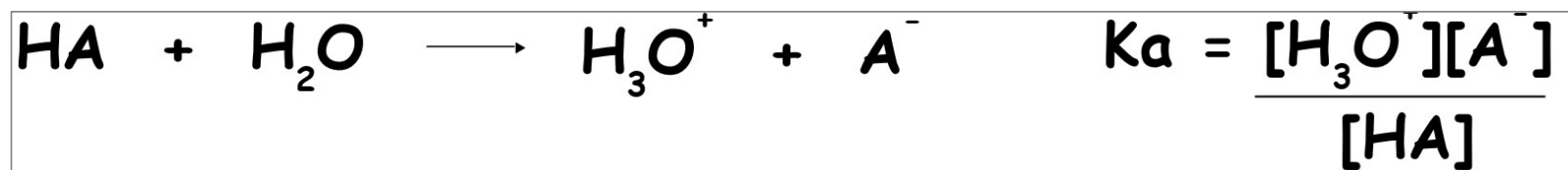
- **Conjugate acid:**
 - The new acid formed when the base gains a proton.
 - always found on the product side.
- **Conjugate base:**
 - The new base formed by removing a proton from an acid.
 - always found on the product side.





Strength of an acid or base

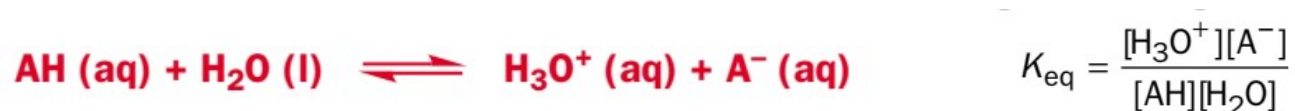
The relative strength of an acid can be determined using the magnitude of K_a (or **pKa**)



- K_a = acid dissociation constant
- K_b = base dissociation constant

The definition of pKa

Now we need to be clearer about ‘strong’ and ‘weak’ acids. In order to measure the strength of an acid relative to water and find out how effective a proton donor it is, we must look at the equilibrium constant for the reaction.



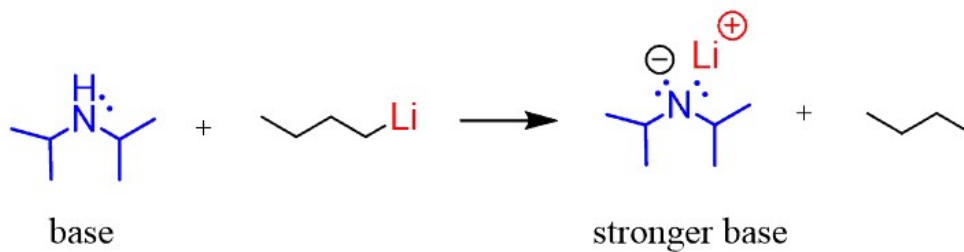
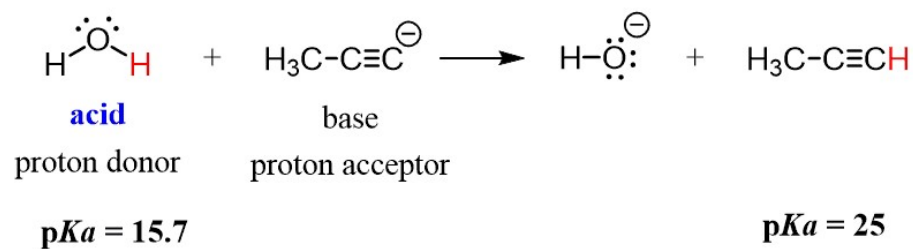
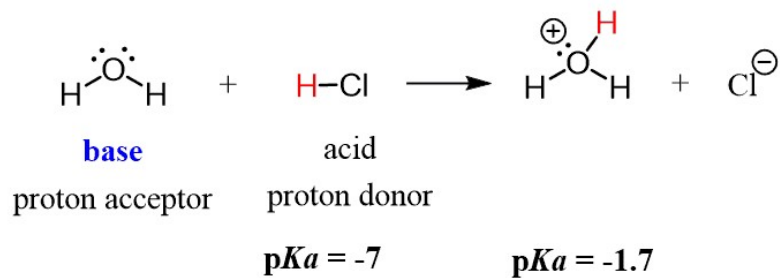
The concentration of water remains constant (at 55.56 mol dm⁻³) with dilute solutions of acids wherever the equilibrium may be and a new equilibrium constant, K_{a} , is defined and called the **acidity constant**.

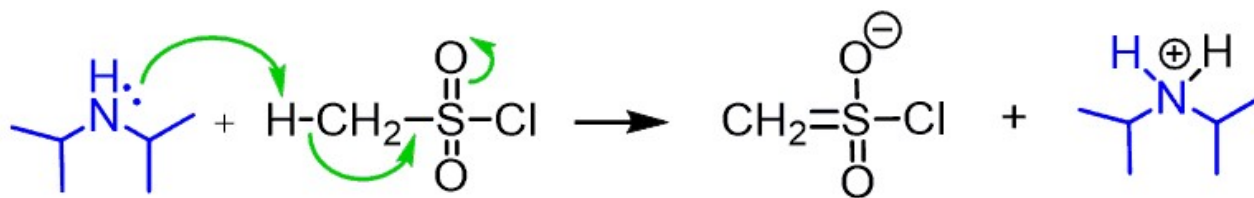
$$K_{\text{a}} = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{AH}]}$$

$$\text{p}K_{\text{a}} = -\log K_{\text{a}}$$

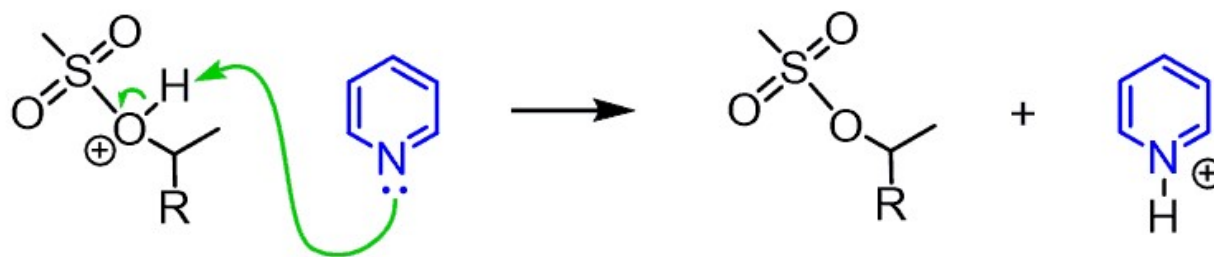
Because of the minus sign in this definition, the lower the $\text{p}K_{\text{a}}$, the larger the equilibrium constant, K_{a} , is and hence the stronger the acid.

Acid-Base role changes depending on the pKa





diisopropylamine
base



Pyridine - base

The pKa value of some compounds

	Functional group	Example	pKa	Conjugate base	
Stronger acid	Hydrochloric acid	HCl	-7	$\text{:}\ddot{\text{Cl}}\text{:}^-$	Weaker base
	Carboxylic acid		5		
	Phenol		10		
	Alcohol		16		
	Ketone		19		
	Alkyne		25		
	Alkene		44		
Weaker acid	Alkane		51		Stronger base

Acid	pK _a	Conjugate base
HI	ca. -10	I ⁻
HCl	ca. -7	Cl ⁻
H ₂ SO ₄	ca. -3	HSO ₄ ⁻
HSO ₄ ⁻	2.0	SO ₄ ²⁻
CH ₃ COOH	4.8	CH ₃ COO ⁻
H ₂ S	7.0	HS ⁻
NH ₄ ⁺	9.2	NH ₃
C ₆ H ₅ OH	10.0	C ₆ H ₅ O ⁻
CH ₃ OH	15.5	CH ₃ O ⁻
	20.0	
	25	
NH ₃	33	NH ₂ ⁻
C ₆ H ₆	ca. 43	C ₆ H ₅ ⁻
CH ₄	ca. 48	CH ₃ ⁻

The relative strength of an acid can be determined using:

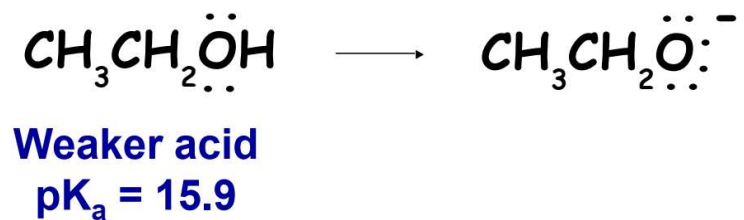
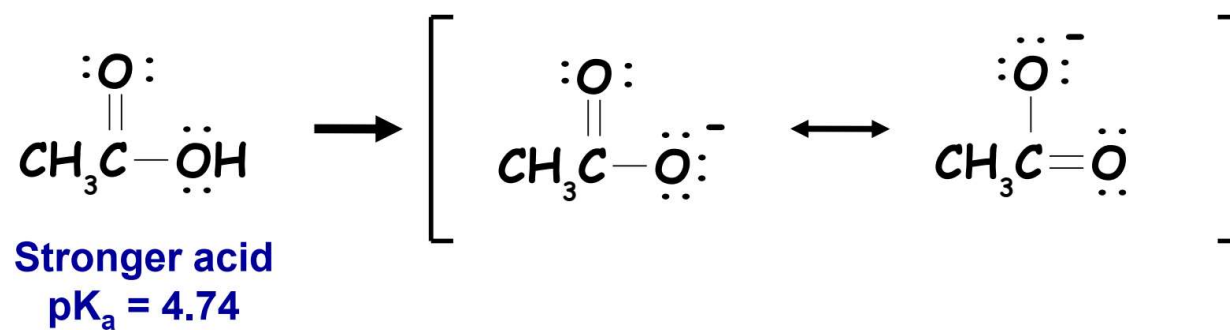
- The magnitude of K_a (or pK_a)
 - As K_a increases, the strength of the acid increases.
 - As pK_a decreases, the strength of the acid increases.
- Structural trend: The strength of an acid, HX, depends on
 - the electronegativity of the atom containing the acidic hydrogen.
 - the stability of the conjugate base, X^-



- Within a group, the strength of an acid increases moving down the group
 - negative charge is more stable when spread out over a larger region.



- Acidity increases when the charge on the conjugate base can be delocalized over two or more atoms through resonance



- The strength of a base is inversely related to its conjugate acid.
- Strong acids form conjugate bases with negligible basicity.



- Weak acids form stronger conjugate bases.

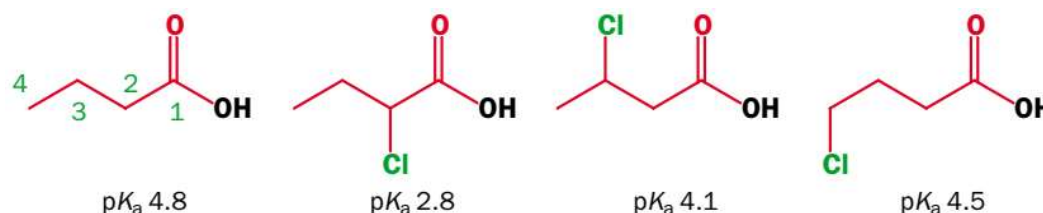


- Substances with negligible acidity form very strong conjugate bases.



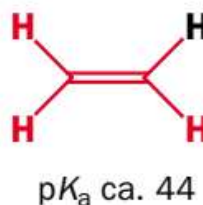
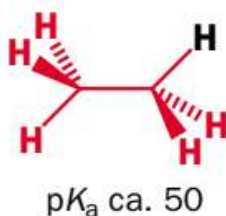
Effect that can affects the pK_a value of some compounds.

Inductive effects:

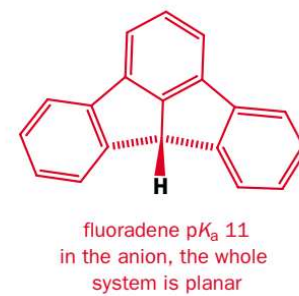
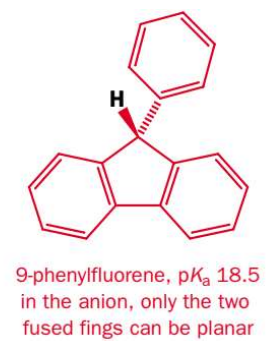
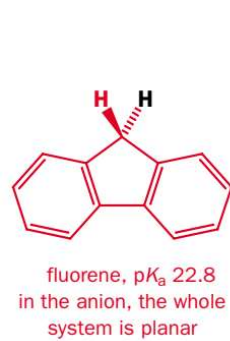
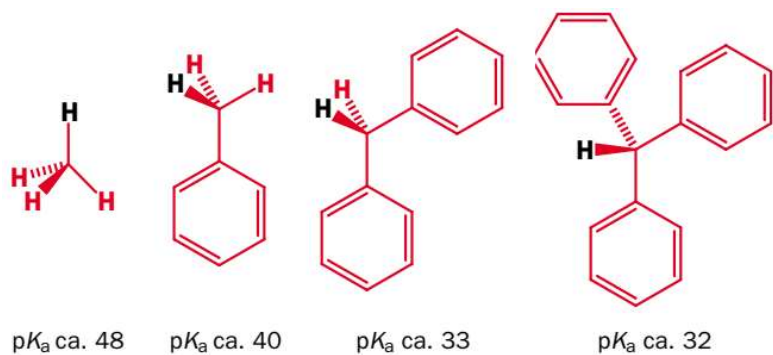


Hybridization:

The hybridization of the orbital from which the proton is removed also affects the pK_a . Since s orbitals are held closer to the nucleus than are p orbitals, so the more s character an orbital has, the more tightly held are the electrons .



Highly conjugated carbon acids:



Aromaticity:



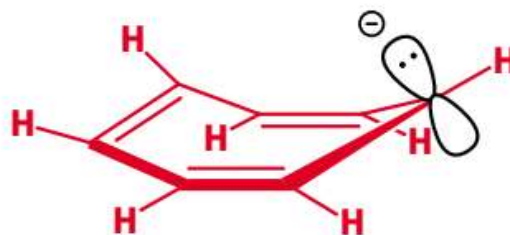
cyclopentadiene
 pK_a 15.5



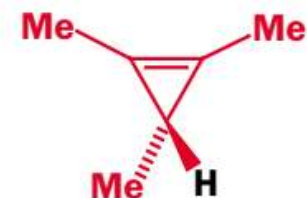
anion is planar
 6π electrons
make it aromatic



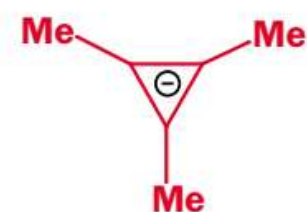
cycloheptatriene
 pK_a ca. 36



anion not planar
neither aromatic or anti-aromatic

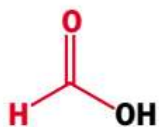


trimethylcyclopropene
 pK_a ca. 62

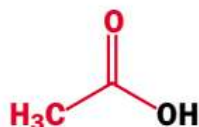


anion is planar
 4π electrons
make it anti-aromatic

Electron-donating groups decrease acidity:



formic (methanoic) acid
 pK_a 3.7



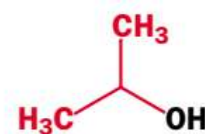
acetic (ethanoic) acid
 pK_a 4.8



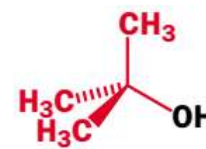
methanol
 pK_a 15.5



ethanol
 pK_a 16.0



isopropyl alcohol
 pK_a 17.1



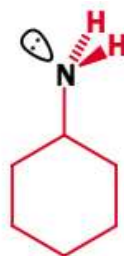
tert-butyl alcohol
 pK_a 19.2

ORGANIC BASES

A **base** is a substance that can accept a proton by donating a pair of electrons.



aniline



Strong bases commonly used in organic chemistry:

potassium *tert*-butoxide: $\text{K}^+\text{OC}(\text{CH}_3)_3$

sodium amide: Na^+NH_2^-

sodium hydride: Na^+H^-

Basicity

How can we measure a base's strength?



ammonia

pK_a ca. 33



ammonium

pK_b 9.24

pK_{aH} of ammonia is **9.24**

we must look up the pK_a of its conjugate acid,
The ammonium cation, NH_4^+

Scales for basicity— pK_b and pK_{aH}



$$K_B = \frac{[\text{OH}^-][\text{BH}^+]}{[\text{B}]}$$

Hence

$$pK_B = -\log(K_B)$$

lower the pK_b , stronger the base

pK_b or pK_{aH} ??

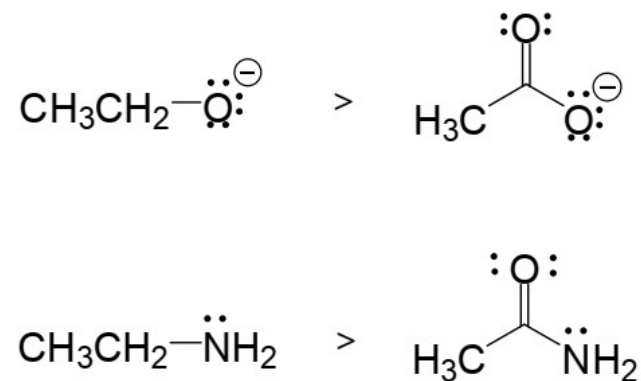
higher the pK_{aH} , stronger the base

What factors affect how basic a compound is?

There are two main factors that determine the strength of a neutral base:

- How accessible is the lone pair.
- What extent can the resultant positive charge formed be stabilized either by delocalization or by the solvent.

Effect of resonance



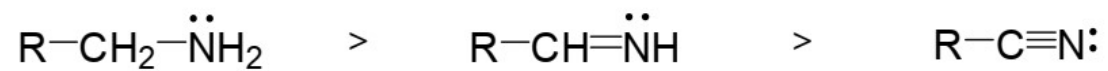
Effect of charge



Effect of electronegativity



Effect of hybridization



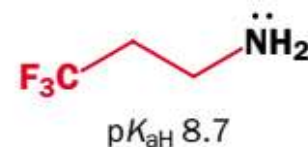
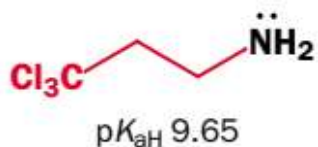
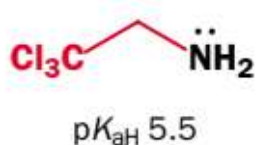
Neutral nitrogen bases

- Ammonia is the simplest nitrogen base and has a pK_aH of 9.24.
- Any substituent that **increases the electron density on the nitrogen** therefore raises the energy of the lone pair thus making it more available for protonation and **increasing the basicity** of the amine .
- Any substituent that **withdraws electron density** from the nitrogen makes it **less basic**.

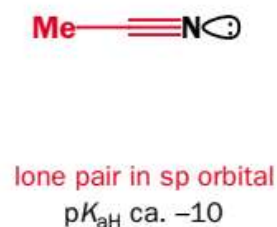
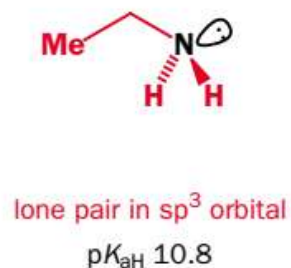
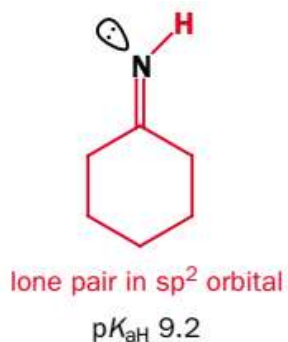
Effects that decrease the electron density on nitrogen

The lone pair on nitrogen will be *less* available for protonation, and the amine *less* basic, if

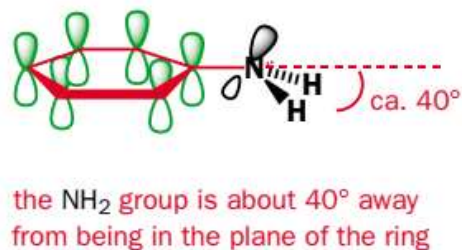
- ✓ The nitrogen atom is attached to an electron-withdrawing group



✓ The lone pair is in an sp or sp_2 hybridized orbital



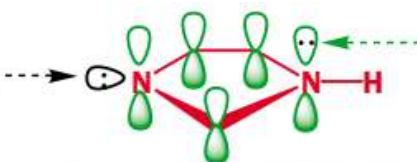
✓ The lone pair is conjugated with an π group



✓ The lone pair is involved in maintaining the aromaticity of the molecule



this lone pair is in an sp^2 orbital and is not involved with the aromaticity of the ring. Protonation occurs here

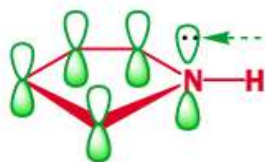


this lone pair is in a p orbital contributing to the 6π electrons in the aromatic ring

the aromaticity of imidazole



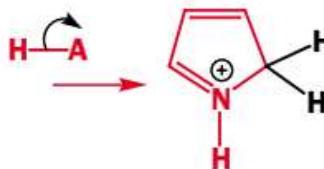
pyrrole
 pK_{aH} ca. -4



the aromaticity of pyrrole



aromatic pyrrole



nonaromatic cation

BTech Attendance
(21th Oct, 1-1.50
pm

